

Markscheme

Mathematics: analysis and approaches
Practice question bank

107. (a)

Consider the expansion of $\left(x^2 + \frac{1}{x}\right)^6$.

Find the term independent of x (the constant term).

$$\text{general term: } \binom{6}{r} x^{2(6-r)} x^{-r} = \binom{6}{r} x^{12-3r} \quad (M1)$$

$$\text{set } 12 - 3r = 0 \Rightarrow r = 4 \quad (M1)$$

$$\binom{6}{4} \quad \mathbf{A1}$$

$$= 15 \quad \mathbf{A1}$$

[4 marks]